Counterfactual Event Bottlenecks for Leakage-Safe Financial Disclosure Modeling

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Abstract

Financial event models often collapse two distinct problems into one representation: ordinary market dynamics and the incremental impact of a newly disclosed event. We propose the Counterfactual Event Bottleneck Transformer (CEBT), an architecture that predicts a no-event baseline from pre-event market history and routes disclosure information through a compact stochastic bottleneck that predicts an event-induced residual. The model is trained with matched same-ticker no-event controls, a residual-suppression loss for controls, and strict temporal feature gates. On a real public-data experiment covering 7,236 SEC 8-K events from 98 usable large U.S. firms between 2020 and 2025, full CEBT does not dominate every baseline on raw regression error, but it is the only tested model with positive abnormal-return rank IC under iid, ticker-clustered, and month-clustered bootstrap intervals. In paired rank-IC comparisons, CEBT significantly improves over no-event dynamics and concat fusion, while its improvement over the no-control CEBT ablation is positive but not statistically resolved. CEBT also suppresses residual magnitude assigned to no-event controls. The main claim is not that CEBT is a deployable trading system, but that an explicit event bottleneck and no-event control objective produce a more temporally disciplined event-residual representation.

1 Introduction

Disclosure events are difficult modeling targets because market movement after a filing is not caused only by the filing. A stock's subsequent return reflects market drift, sector movement, liquidity, firm-specific momentum, and the information content of the disclosure itself. Standard fusion models mix these sources into one hidden representation, making it hard to tell whether the model learned an event effect or simply reproduced ordinary dynamics.

We introduce CEBT, an event modeling architecture that decomposes the forecast into

$$\hat{y}_{t+h} = f_{\text{no event}}(x_{<t}) + g_{\text{event}}(z_e, x_{<t}),$$

where $f_{\text{no event}}$ is learned from pre-event numeric history and z_e is a compact stochastic disclosure bottleneck. The bottleneck is trained with matched no-event controls: dates for the same ticker that are far from known SEC 8-K events. For controls, the model is penalized when it assigns a large event residual. This makes the residual interpretable as a candidate event-specific increment rather than an unconstrained shortcut.

Our contributions are:

1. A no-event-plus-residual neural architecture for financial disclosure modeling.
2. A stochastic event bottleneck trained with matched no-event controls.
3. A leakage-safe public-data implementation using SEC accepted timestamps and public price histories.
4. A real-data study showing that residual control changes model behavior and that CEBT produces the strongest abnormal-return rank ordering under clustered uncertainty checks.

2 Related Work

CEBT is closest to causal and event-driven financial forecasting. Classical event-study methods estimate abnormal returns around public information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]; CEBT keeps this event-study grounding but learns a neural decomposition of baseline dynamics and disclosure residuals. CAMEF studies causal-augmented multi-modal financial forecasting [Chen et al., 2025]. Causal Transformer and G-Transformer estimate counterfactual outcomes in temporal settings [Melnychuk et al., 2022, Feng et al., 2024]. The architectural ingredients also connect to Transformers [Vaswani et al., 2017], variational latent-variable models [Kingma and Welling, 2013], and the information bottleneck principle [Tishby et al., 2000]. Time-series foundation models such as Chronos, Time-LLM, MOMENT, and Mamba improve sequence modeling capacity but do not isolate disclosure-induced residuals [Ansari et al., 2024, Jin et al., 2023, Wu et al., 2024, Gu and Dao, 2023]. Leakage-focused work such as Profit Mirage motivates strict temporal evaluation in financial agents [Li et al., 2025]. Subliminal-learning results further motivate caution around hidden outcome signal in generated or compressed representations [Cloud et al., 2025].

3 Method

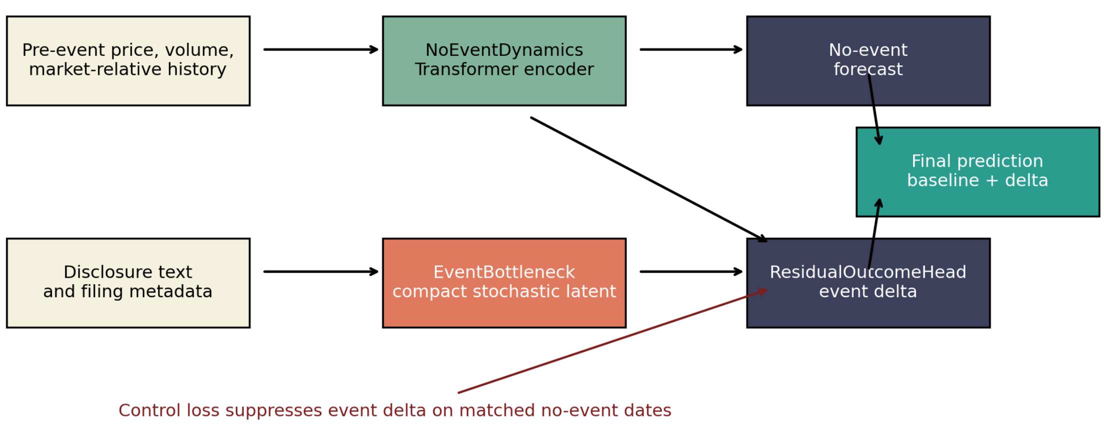

Figure 1: CEBT decomposes a forecast into no-event dynamics and an event-induced residual routed through a compact stochastic bottleneck. Matched no-event controls penalize residuals on dates where no disclosure event should be needed.

NoEventDynamics. The no-event module receives a fixed pre-event window of price, volume, market, and relative-return features. A Transformer encoder pools this history and predicts the ordinary forward movement expected without relying on the event text.

EventBottleneck. The event module embeds the disclosure text and metadata, then parameterizes a Gaussian latent $z_e \sim \mathcal{N}(\mu_e, \sigma_e^2)$. The latent dimension is intentionally small. We add a KL term and sparsity term so the event representation cannot behave like an unrestricted text embedding.

ResidualOutcomeHead. The residual head predicts an event-induced delta from the no-event state, metadata, and z_e. The final forecast is the sum of the no-event forecast and this residual.

Training objective. The loss combines supervised MSE, a no-event control residual penalty, KL regularization, latent sparsity, and a small residual-variance consistency penalty:

$$\mathcal{L} = \lambda_y \|\hat{y} - y\|_2^2 + \lambda_c \mathbb{1}_{control} \|\Delta_e\|_2^2 + \lambda_{kl} D_{KL}(q(z_e) \| p(z)) + \lambda_s \|z_e\|_1 + \lambda_v \mathrm{Var}(\Delta_e).$$

4 Data and Leakage Controls

We build a public-data experiment from SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The paper-scale run covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and daily public prices. Each event row stores CIK, ticker, accession number, form type, filing date, accepted timestamp, available timestamp, source URL, text hash, and local provenance. Event label windows start from the first tradable session after public availability. If a filing is accepted after 16:00 ET, the label starts on the next trading day.

Features use only prices strictly before the event label start date. Labels use future returns only as targets. We construct one same-ticker no-event control for each filing, excluding dates within a 10-calendar-day blackout window around known events.

Table 1: Paper-scale public-data substrate.

Artifact	Count
Usable companies	98
SEC 8-K events	7,236
Matched no-event controls	7,236
Feature rows	14,472
Public price rows	159,093
Held-out test rows	2,463

5 Experiments

We compare five models: no-event dynamics only, text-only MLP, concat-fusion, CEBT without control losses, and full CEBT. The target vector contains five-trading-day abnormal return, volatility jump, and volume jump. We report iid row bootstrap intervals and clustered bootstrap sensitivity intervals by ticker and by label-start month.

Table 2: Held-out paper-scale results. MSE is over the three target dimensions. Rank-IC intervals shown here are ticker-clustered; month-clustered intervals are discussed in the text.

Model	MSE	95% CI	Rank IC	Ticker-cluster 95% CI
No-event	0.0696	[0.0535, 0.0907]	-0.0411	[-0.0840, 0.0016]
Text-only	0.0643	[0.0512, 0.0820]	-0.0009	[-0.0456, 0.0434]
Concat fusion	**0.0597**	[0.0456, 0.0784]	-0.0182	[-0.0597, 0.0267]
CEBT, no controls	0.0666	[0.0507, 0.0874]	0.0279	[-0.0139, 0.0716]
CEBT	0.0660	[0.0504, 0.0867]	**0.0463**	[0.0052, 0.0893]

Table 3: Paired MSE comparisons against full CEBT. Positive values mean CEBT has lower MSE than the baseline.

Baseline	Baseline–CEBT MSE	95% CI	CEBT better?
CEBT, no controls	0.00056	[0.00019, 0.00097]	Yes
Concat fusion	-0.00630	[-0.00863, -0.00416]	No
No-event	0.00359	[0.00184, 0.00549]	Yes
Text-only	-0.00171	[-0.00524, 0.00143]	No

Table 4: Paired abnormal-return rank-IC comparisons against full CEBT. Positive values mean CEBT has higher rank IC than the baseline.

Baseline	CEBT–baseline rank IC	95% CI	CEBT better?
CEBT, no controls	0.0184	[-0.0202, 0.0595]	No
Concat fusion	0.0645	[0.0058, 0.1215]	Yes
No-event	0.0874	[0.0251, 0.1534]	Yes
Text-only	0.0472	[-0.0097, 0.1028]	No

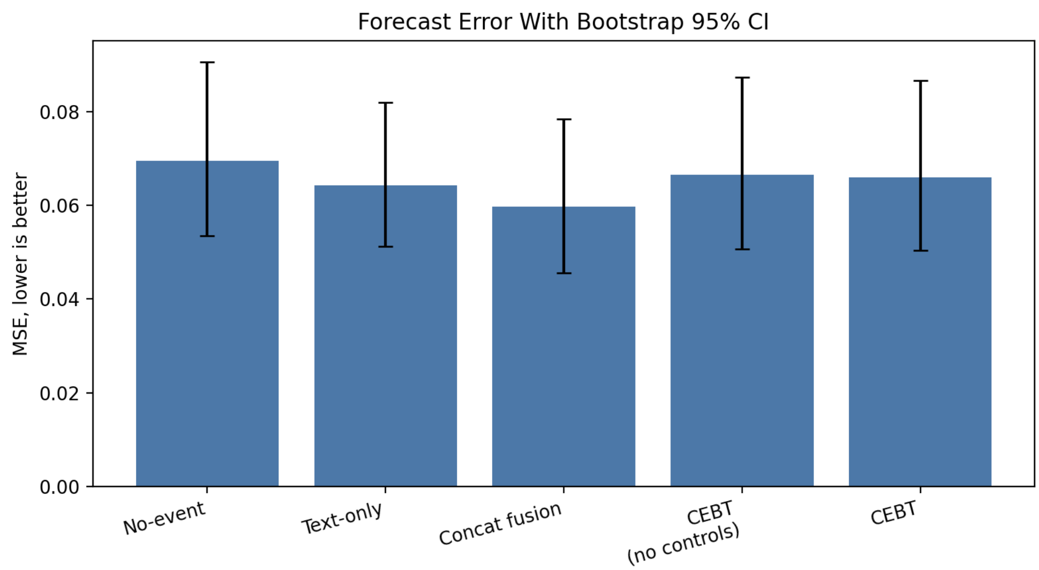

Figure 2: Forecast error with bootstrap confidence intervals. Concat fusion has the strongest point-estimate MSE; CEBT's advantage appears in rank ordering and residual discipline rather than raw multi-target regression error.

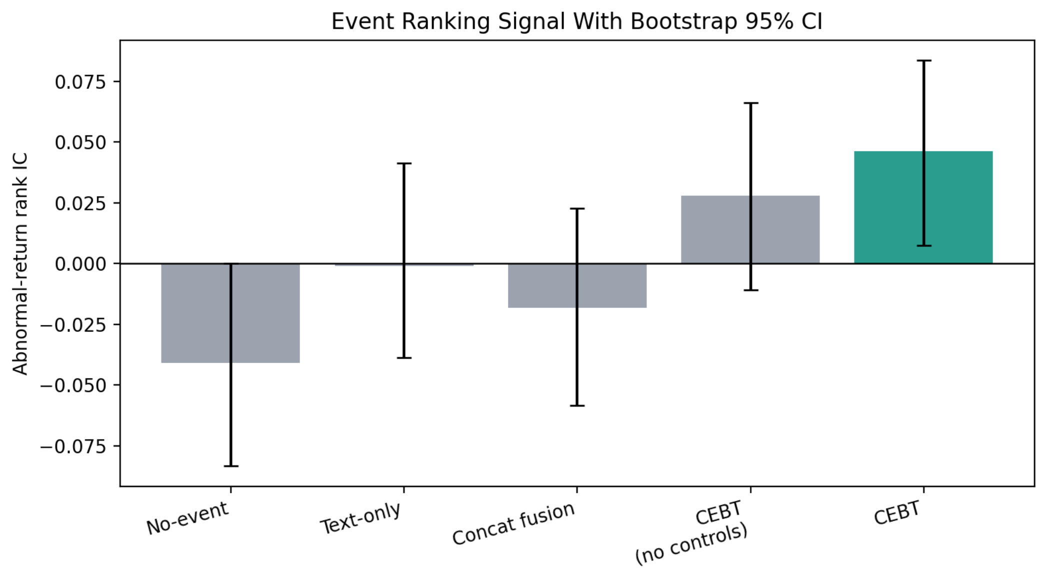

Figure 3: Abnormal-return rank IC with bootstrap confidence intervals. Full CEBT is the only tested model whose interval is entirely above zero.

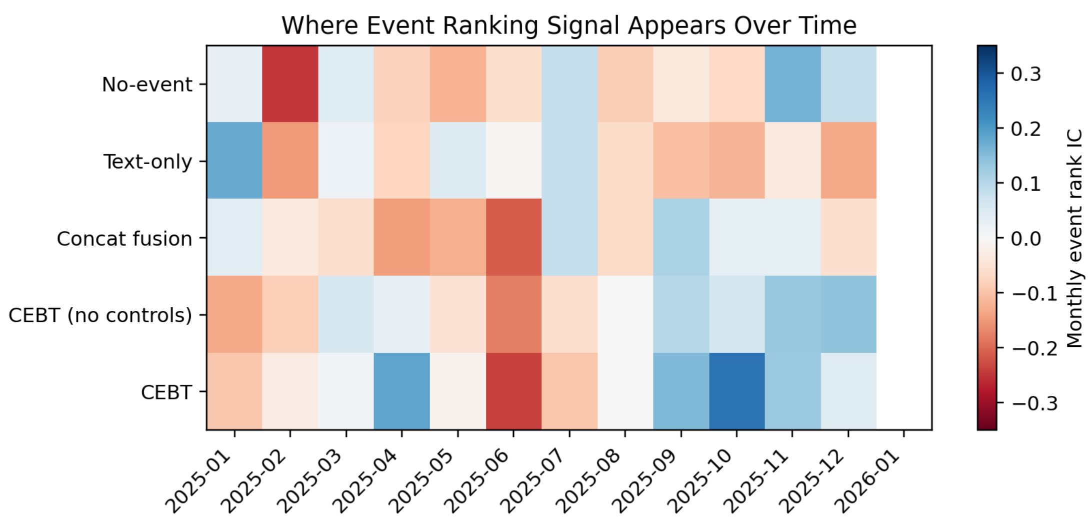

Figure 4: Monthly event-ranking signal by model. The heatmap shows where rank signal appears and whether it is temporally concentrated or persistent across the held-out period.

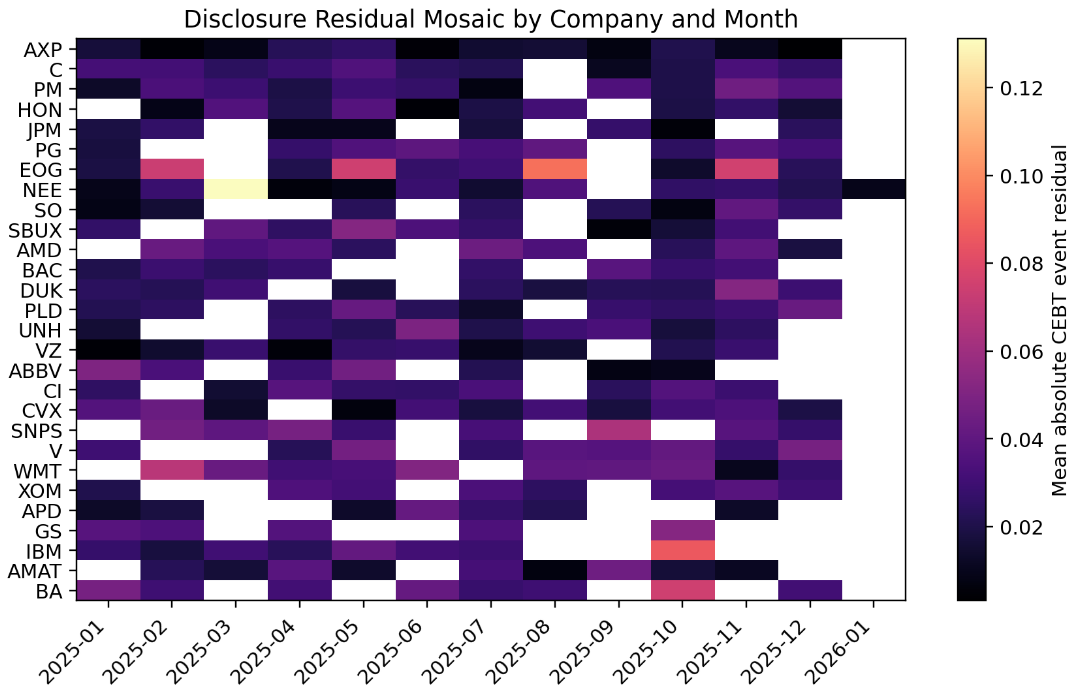

Figure 5: CEBT event-residual mosaic by company and month. Darker cells indicate months where the model routes more mass through the event residual rather than the no-event baseline; white cells indicate no held-out real 8-K events for that company-month.

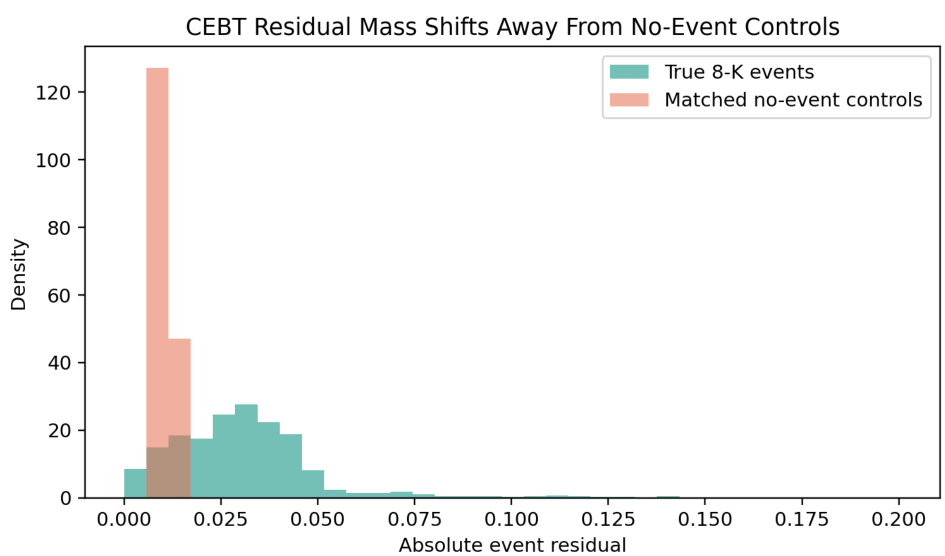

Figure 6: Residual distribution for true 8-K events and matched no-event controls. The distributional view is more informative than a mean-only bar chart: controls concentrate near zero, while true events have a heavier residual tail.

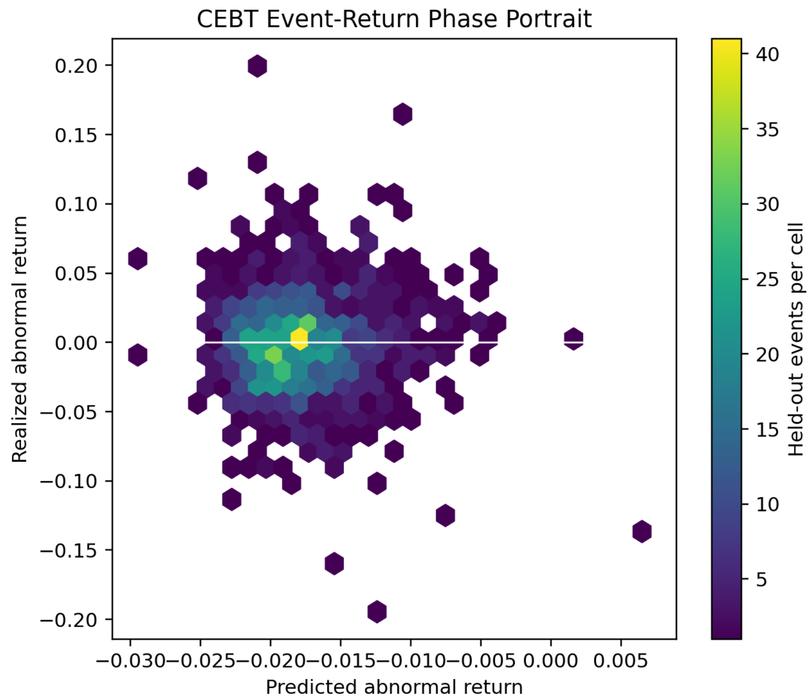

Figure 7: CEBT event-return phase portrait on held-out real disclosures. The density plot shows where predictions and realized abnormal returns concentrate, making the narrow prediction band and wide realized-return dispersion visible without plotting thousands of overlapping points.

6 Findings

CEBT is an event-ranking model, not the best raw regressor. Concat fusion has the best point-estimate MSE, 0.0597, while full CEBT has MSE 0.0660. This matters: the architectural decomposition does not automatically minimize multi-target regression error. The evidence for CEBT is instead concentrated in rank ordering of event outcomes.

The event-ranking signal survives clustered uncertainty checks. Full CEBT has abnormal-return rank IC 0.0463. Its iid bootstrap interval is [0.0074, 0.0836], the ticker-clustered interval is [0.0052, 0.0893], and the month-clustered interval is [0.0112, 0.0768]. No other tested model has a positive clustered interval excluding zero.

Paired rank comparisons separate CEBT from stronger baselines. CEBT improves paired rank IC over concat fusion by 0.0645 with a 95% paired interval [0.0058, 0.1215], and over no-event dynamics by 0.0874 with interval [0.0251, 0.1534]. The improvement over CEBT without controls is positive, 0.0184, but not resolved by the paired interval [-0.0202, 0.0595].

Control losses alter the residual semantics. The full CEBT model assigns mean absolute event residual 0.0313 to true events and 0.0103 to controls. The no-control variant assigns 0.0396 to true events and 0.0298 to controls. Thus, the control objective reduces spurious residual assignment to no-event dates by about 66%, while preserving a roughly 3.0x event-to-control residual ratio.

7 Limitations

This is a first paper-scale run, not a final investment model. The text encoder is a deterministic open hashing baseline rather than a large pretrained financial encoder. The public price source is not CRSP, and labels use daily closes rather than intraday event windows. The universe is restricted to 8-K filings, and two configured tickers did not yield final usable event rows because of SEC retrieval failures during the build. The next version should test pretrained open embeddings, 10-Q/10-K extensions, sector controls, and harder placebo controls where no-event dates receive unrelated historical disclosure text instead of a zero event embedding.

8 Conclusion

CEBT reframes event-driven financial modeling as a decomposition problem: ordinary market dynamics plus a disclosure-induced residual. In a real public-data experiment, CEBT does not beat concat fusion on raw MSE, but it produces the strongest abnormal-return rank ordering and remains positive under ticker- and month-clustered uncertainty checks. The result supports CEBT as a promising architecture for studying event-specific effects under strict temporal information boundaries.

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